MILITARY STANDARD

RAILCAR LOADING OF HAZARDOUS MATERIALS

PROJECTILES 5/54
IN PALLET ADAPTER MK 11 AND MODS
FLEET ISSUE UNIT LOAD

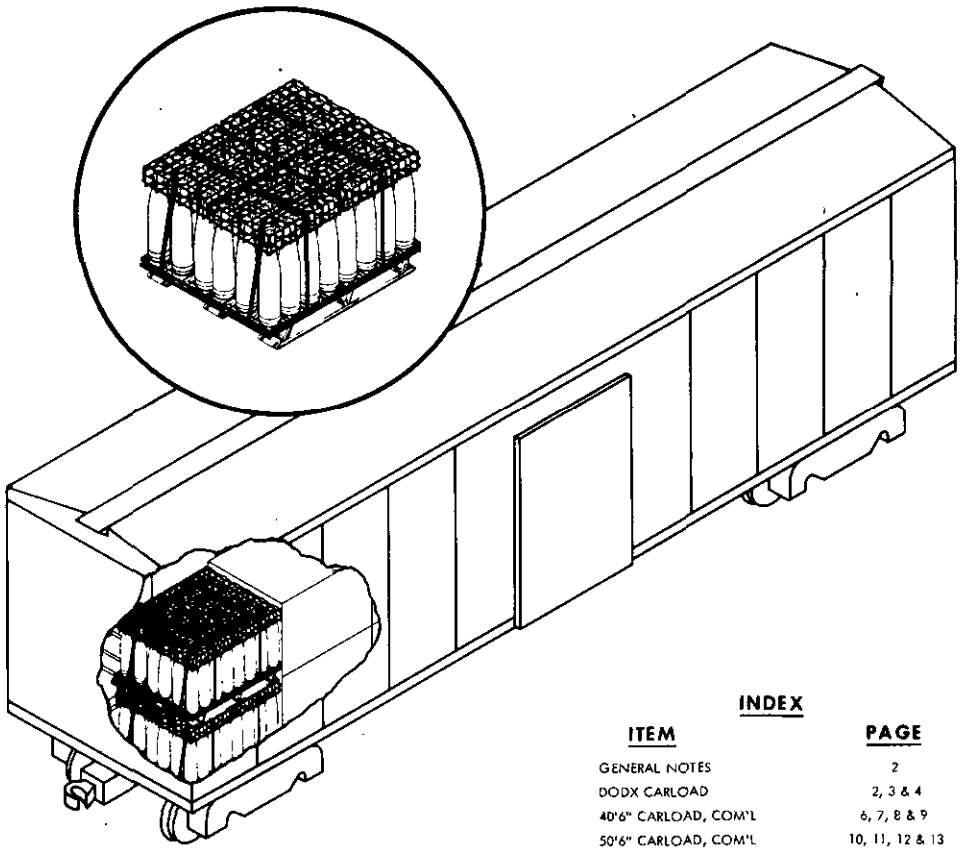
MIL-STD 1325-15
(NAVY)

22 DEC 1975

SUPERSEDING WR 52/15
23 MARCH 1965

UNIT LOAD DATA

UNIT LOAD DRAWINGS ..... WR-54/6 AND WR-54/271
WEIGHT ..... SEE TABLE I
DIMENSIONS ..... SEE TABLE I
CUBE ..... SEE TABLE I
HAZARD CLASSIFICATION ..... SEE TABLE II



INDEX

ITEM	PAGE
GENERAL NOTES	2
DODX CARLOAD	2, 3 & 4
40'6" CARLOAD, COM'L	6, 7, 8 & 9
50'6" CARLOAD, COM'L	10, 11, 12 & 13
DUMMY UNIT LOAD	14

NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCK-LOADING AND CONTAINERLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY).

FSC 8140

AUTHORIZED AND RELEASED FOR GENERAL USE.

M.S. Gray NWHL 12/16/75
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J.E. Kelly 12/17/75
SIGNATURE SEASYSCOM BY DIRECTION DATE

APPROVED BY BUREAU OF EXPLOSIVES

A.J. Groosmuck 10/28/75
SIGNATURE SUPERVISOR, MILITARY & INTERMODEL SERVICES DATE

ORIGINATOR L.P. Cavanaugh 12/15/75
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.

GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS MATERIALS".
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT. UTILITY LOADER CARS SHALL BE SERIES DODX 29000.
3. LOADING PLANS SHOWN ARE FOR DODX UTILITY LOADER CAR WITH 50 FT 6 INCHES INSIDE LENGTH, 107 3/4 INCHES INSIDE WIDTH BETWEEN RAILS (111 INCHES INSIDE WIDTH BETWEEN SIDE WALLS), COMMERCIAL BOXCARS WITH 40 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH, AND COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH.
4. THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. THE LOAD CONSISTS OF 5"/54 PROJECTILES PALLETIZED IN ACCORDANCE WITH WR-54/6 OR WR-54/271.
7. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY).
9. ALL STRAP JOINTS SHALL BE CRIMP-TYPE JOINTS MADE WITH A CRIMPING TOOL THAT DOES NOT CUT THE SEAL OR STRAP, RATHER THAN A TOOL THAT MAKES A NOTCH-TYPE JOINT. THE STRAPPING MUST BE IN ACCORDANCE WITH NAVSUP NOTICE 4642 DATED 21 APRIL 1971.
10. APPLICABLE MATERIAL SPECIFICATIONS:
    DUNNAGE LUMBER - FED. SPEC MM-L-751
    NAILS - FED. SPEC FF-N-105
    STRAPPING - FED. SPEC QQ-S-781, TYPE I, HEAVY DUTY, CLASS A, DRY (UNLUBRICATED)
    SEALS - FED. SPEC QQ-S-781, STYLE III, HEAVY DUTY
11. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA, CLOSE AND SEAL BOXCAR DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED IN OP 2165 (VOL. 1).

50 FT 6 IN. BOXCAR, DODX

1. THE CARLOAD CONSISTS OF 28 OR 30 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. A DETAILED DESCRIPTION AND OPERATING INSTRUCTIONS FOR THE UTILITY LOADER ARE CONTAINED IN OP 1750.
3. TO PREVENT UNUSED "DF" EQUIPMENT FROM BECOMING DISLODGED DURING TRANSIT OF DODX CARS SECURE IT AT ANY LOCATION IN THE BOXCAR WHICH WILL NOT INTERFERE WITH UNLOADING.
4. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN DODX BOXCARS THE SAME PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE. ANY BAYS OR PORTION THEREOF MAY BE USED PROVIDING THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR ARE COMPLIED WITH (SEE MIL-STD-1325 (NAVY)). EACH CROSSMEMBER WILL BE USED IN SUCH A MANNER THAT IT WILL RETAIN NOT MORE THAN 2000 LBS OF THE LADING.

TABLE I

TYPE OF PROJECTILE	UNIT LOAD DIMENSIONS L W H	UNIT LOAD WT (LBS) * PALLET ADAPTER MK 11 MOD 0	MOD 1	CUBE CU FT
PROJECTILE W/O WATERPROOF PROTECTING CAP	40 x 48 x 31 3/4	3683	3627	35.2
PROJECTILE W/WATERPROOF PROTECTING CAP	40 x 48 x 33 3/4	3779	3723	37.5
ROCKET ASSISTED PROJECTILE MK 58 MOD 0	40 x 48 x 34	3443	3387	37.9
PROJECTILEW/PLASTIC ROTATING BAND (HI-FRAG)	40 x 48 x 34 3/4	3632 **		38.6

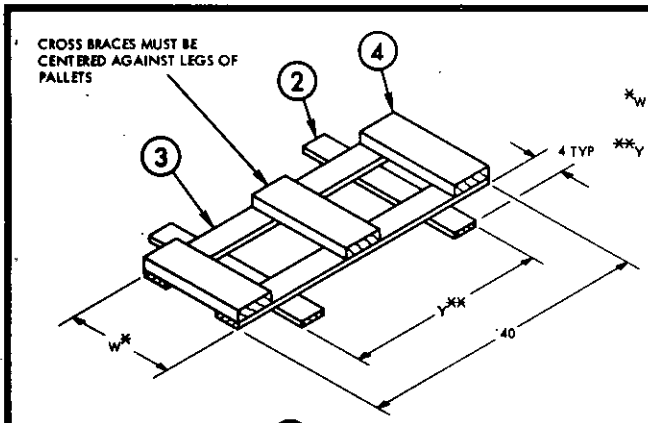
* WEIGHTS ARE APPROXIMATE

** PALLET ADAPTER MK 11 MOD 2 BOTTOM AND MK 11 MOD 1 TOP

TABLE II

HAZARD CLASSIFICATION	
TYPE OF PROJECTILE	DOT
TP	B
HE-CVT	A
CHAFF	B
SMK-WP	A
VT	A
AAC	A
HC	A
COM	A
IR	A
ILLUM	B
HE-PD	A
VT-NF	B
IR-NF	B
HI-FRAG	A

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.



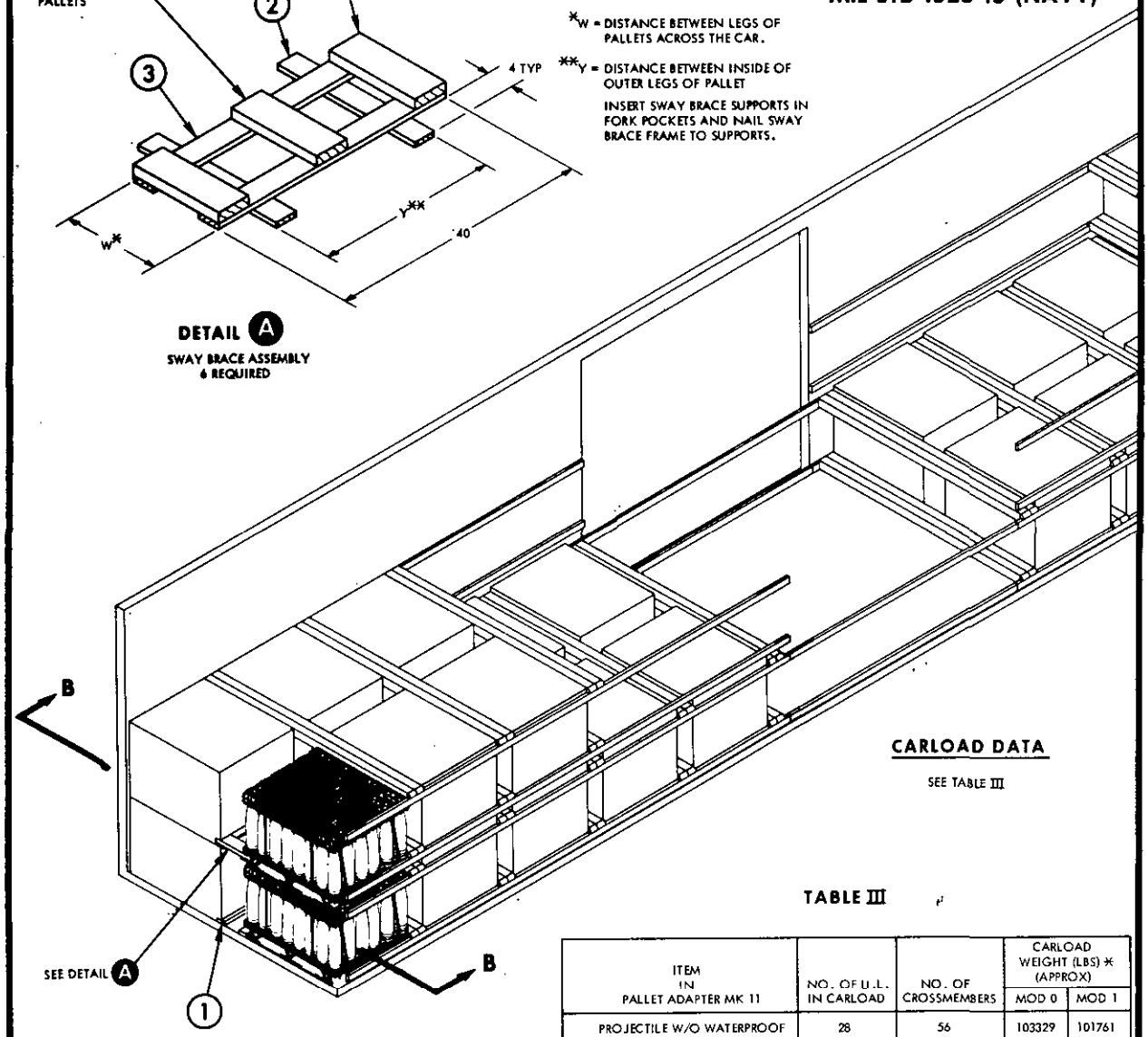
*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

**Y = DISTANCE BETWEEN INSIDE OF OUTER LEGS OF PALLET

INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.

DETAIL A

SWAY BRACE ASSEMBLY
6 REQUIRED



CARLOAD DATA

SEE TABLE III

TABLE III

ITEM IN PALLET ADAPTER MK 11	NO. OF U.L. IN CARLOAD	NO. OF CROSSMEMBERS	CARLOAD WEIGHT (LBS) * (APPROX)	
			MOD 0	MOD 1
PROJECTILE W/O WATERPROOF PROTECTING CAP	28	56	103329	101761
PROJECTILE W/WATERPROOF PROTECTING CAP	28	56	106017	104429
ROCKET ASSISTED PROJECTILE MK 58 MOD 0	30	60	103495	101815
** PROJECTILE W/PLASTIC ROTATING BAND (HI-FRAG)	28	56	101901	

* INCLUDES 205 LBS (APPROX) DUNNAGE WEIGHT

** UNIT LOAD HAS PALLET ADAPTER MK 11 MOD 2 BOTTOM AND MK 11 MOD 1 TOP

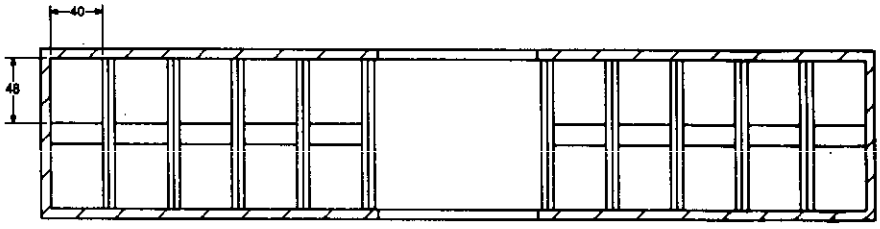
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS: NUMBER	NAILS: SIZE
4	CROSS BRACE	2 x 6 x W *	15	SEE 3	-	-
3	SWAY BRACE LONGITUDINAL	1 x 4 x 40	10	2, 4	SEE NOTE **	
2	SWAY BRACE SUPPORT	1 x 4 x (W + 8) *	10	SEE 3	-	-
1	SLEEPER	2 x 4 x 40	20	CAR FLOOR	2 PER FOOT	16d

LIST OF MATERIALS AND NAILING DATA

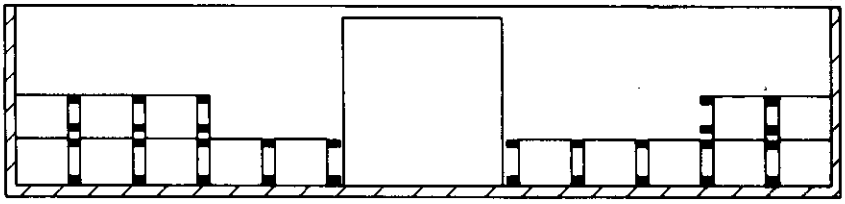
* W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR

** PIECE 3 NAILS TO PIECE 4 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 2 WITH 6d NAILS, 1 PER JOINT

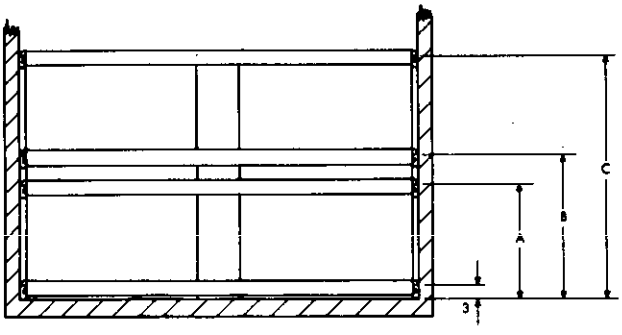
Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.



LOADING PLAN



LOADING PLAN
ELEVATION



SECTION B-B

SHOWING LOCATION OF WALL MEMBERS
USED FOR CROSSMEMBERS
SEE TABLE IV

TABLE IV

ITEM	A	B	C
PROJECTILE W/O WATERPROOF PROTECTING CAP	32	38	64
PROJECTILE W/WATERPROOF PROTECTING CAP	32	40	68
ROCKET ASSISTED PROJECTILE MK 58 MOD 0	32	40	68
PROJECTILE W/PLASTIC ROTATING BAND (HI-FRAG)	32	40	68

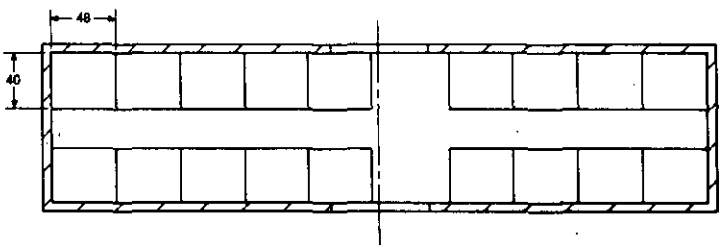
Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY LEFT BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.

40 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD MAY CONSIST OF 36 UNIT LOADS WITH 18 IN THE FIRST LAYER AND 18 IN THE SECOND LAYER (48" DIMENSION OF U.L. LENGTHWISE IN THE CAR), OR 34 UNIT LOADS BY SUBSTITUTING DUMMY UNIT LOADS IN THE SECOND LAYER OF THE DOORWAY STACK. THE CARLOAD MAY ALSO CONSIST OF 22 UNIT LOADS IN A SINGLE LAYER (48" DIMENSION OF U.L. CROSSWISE IN THE CAR), OR A SINGLE LAYER OF 22 UNIT LOADS WITH A PARTIAL UPPER LAYER OF UP TO 12 UNIT LOADS (UPPER PARTIAL LAYER DIVIDED APPROXIMATELY EQUALLY EACH END OF THE CAR). THE PARTIAL LAYER SHALL BE DUNNAGED BY PARTIAL LAYER BRACING IN ACCORDANCE WITH THE PROCEDURES DEPICTED IN MIL-STD-1325-100 (NAVY) AND/OR MIL-STD-1325-102 (NAVY). THE NUMBER OF UNIT LOADS IN THE CAR DEPENDS ON THE QUANTITY TO BE SHIPPED AND THE LOAD LIMIT OF THE BOXCAR.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH, SEE MIL-STD-1325 (NAVY).
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT WIDE PROVIDED DOORWAY PROTECTION WHEN REQUIRED IS INSTALLED IN ACCORDANCE WITH MIL-STD-1325 (NAVY).



LOADING PLAN

* USE SWAY BRACE ASSEMBLY DETAIL A FOR STEEL FLOOR CARS

** W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR

*** PIECE 3 NAILS TO PIECE 4 WITH 10d NAILS, CLINCHED, 3 PER JOINT AND TO PIECE 2 WITH 6d NAILS, 1 PER JOINT

**** 2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS: NUMBER	NAILS: SIZE
	LIST OF MATERIALS AND NAILING DATA					
13	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	8	12	2 PER JOINT	10d
12	VERTICAL DOORWAY MEMBER	2 x 3 x 90	4	DOOR POST	2 PER FOOT	20d
11	STRUT	4 x 4 x WEDGE FIT ****	32	6	2 PER JOINT	16d
10	HOLD DOWN CLEAT	2 x 3 x CUT TO FIT	4	9	5	16d
9	HOLD DOWN CLEAT	2 x 3 x CUT TO FIT	4	8	5	10d
8	HOLD DOWN SPACER	2 x 4 x CAR WIDTH-1	2	6	3 PER JOINT	10d
7	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	6	3 PER JOINT	10d
6	CENTER GATE VERTICAL	2 x 6 x 72	16	SEE 5	-	-
5	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	8	6	3 PER JOINT	10d
4	CROSS BRACE	2 x 6 x W **	27	SEE 3	-	-
3	SWAY BRACE LONGITUDINAL	1 x 4 x 48	18	2, 4	SEE NOTE ***	
2	SWAY BRACE SUPPORT	1 x 4 x (W+8) **	18	SEE 3	-	-
1	SLEEPER	2 x 4 x 48 *	18	CAR FLOOR	2 PER FOOT	16d

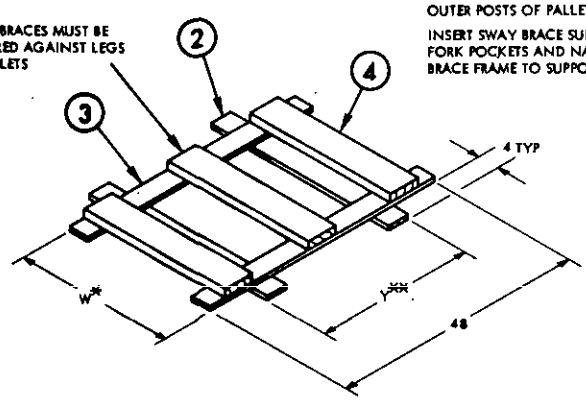
Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.

*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

**Y = DISTANCE BETWEEN INSIDE OF OUTER POSTS OF PALLET.

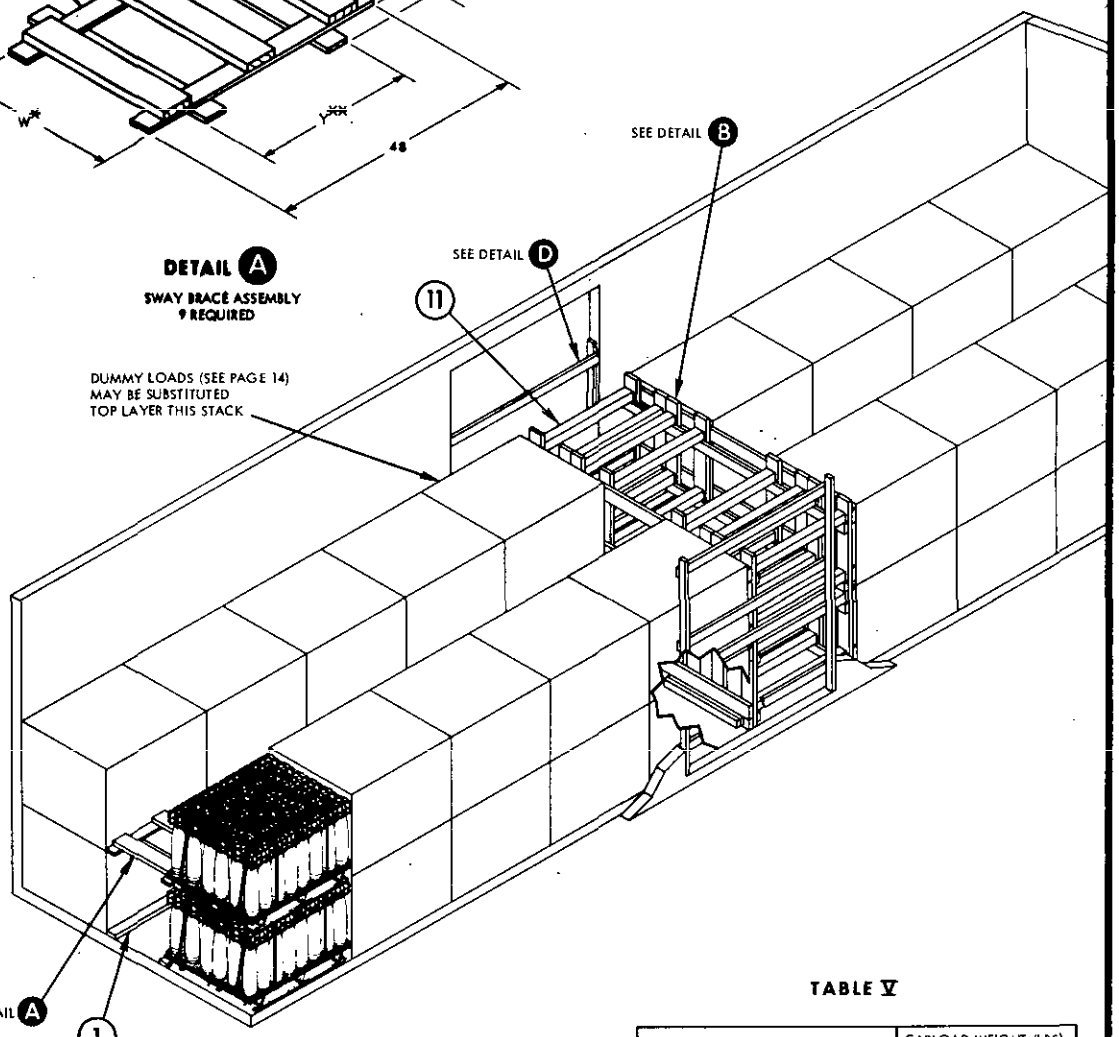
INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.

CROSS BRACES MUST BE CENTERED AGAINST LEGS OF PALLETS



DETAIL A

SWAY BRACE ASSEMBLY 9 REQUIRED



USE SWAY BRACE ASSEMBLY DETAIL A FOR STEEL FLOOR CARS

CARLOAD DATA

NUMBER OF UNIT LOADS 36

CARLOAD WEIGHT ..... SEE TABLE V

TABLE V

ITEM	CARLOAD WEIGHT (LBS) (INCLUDES LBS APPROX DUNNAGE WT)* PALLET ADAPTER MK 11	
	MOD 0	MOD 1
PROJECTILE W/O WATERPROOF PROTECTING CAP	133850	131834
PROJECTILE W/WATERPROOF PROTECTING CAP	137306	135290
ROCKET ASSISTED PROJECTILE MK 58 MOD 0	125210	123194
** PROJECTILE W/PLASTIC ROTATING BAND (HI-FRAG)	132014	

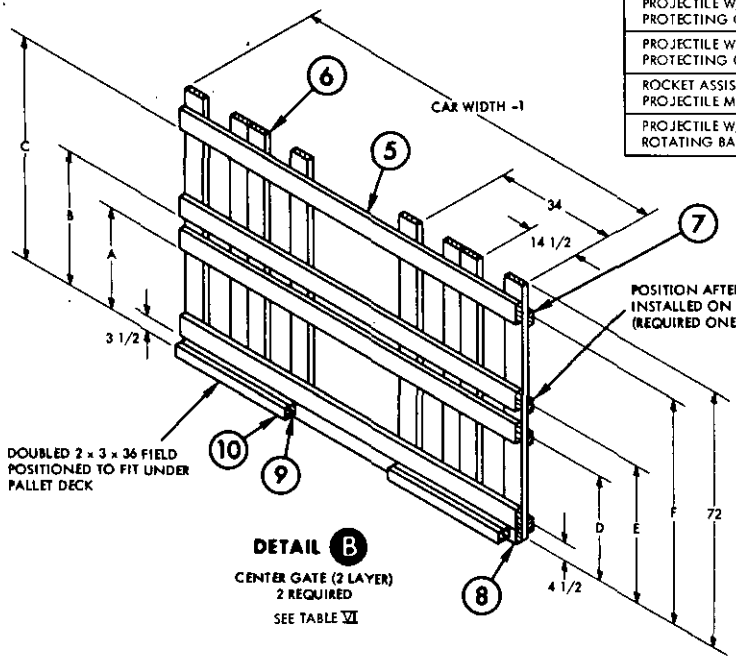
* APPROXIMATE DUNNAGE WT = 1262 LBS (APPROX)

** UNIT LOAD HAS PALLET ADAPTER MK 11 MOD 2 BOTTOM AND MK 11 MOD 1 TOP

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.

TABLE VI

ITEM	A	B	C	D	E	F
PROJECTILE W/O WATERPROOF PROTECTING CAP	26 1/4	35 1/4	58	27 1/4	36 1/4	59
PROJECTILE W/WATERPROOF PROTECTING CAP	28 1/4	37 1/4	62	29 1/4	38 1/4	63
ROCKET ASSISTED PROJECTILE MK 58 MOD 0	28 1/2	37 1/2	62 1/2	29 1/2	38 1/2	63 1/2
PROJECTILE W/PLASTIC ROTATING BAND (HI-FRAG)	29 1/4	38 1/4	64	30 1/4	39 1/4	65



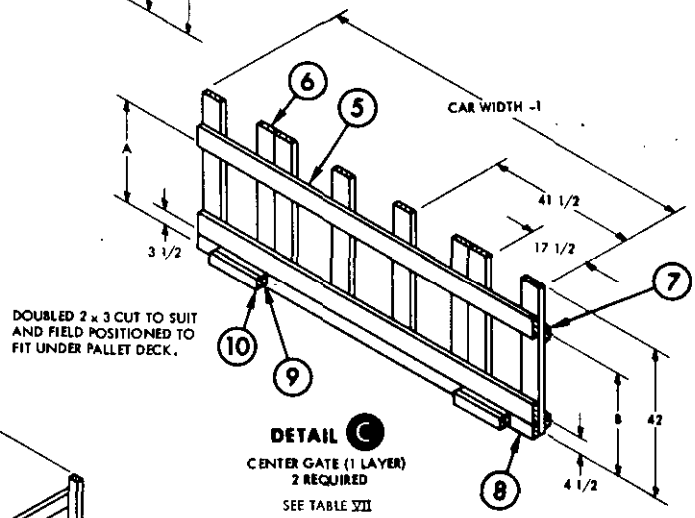
DETAIL B

CENTER GATE (2 LAYER)
2 REQUIRED

SEE TABLE VI

TABLE VII

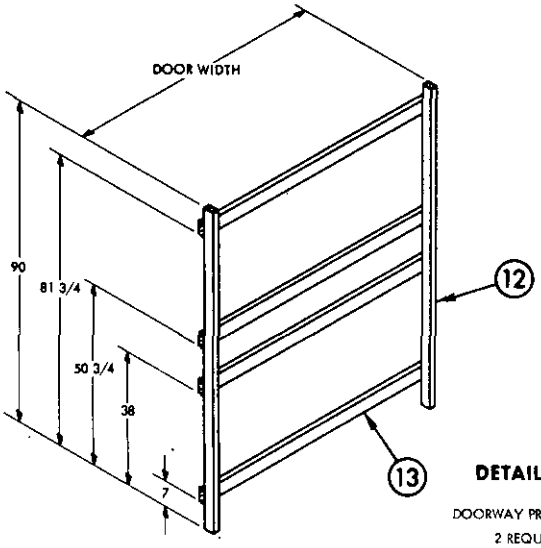
ITEM	A	B
PROJECTILE W/O WATERPROOF PROTECTING CAP	26 1/4	27 1/4
PROJECTILE W/WATERPROOF PROTECTING CAP	28 1/4	29 1/4
ROCKET ASSISTED PROJECTILE MK 58 MOD 0	28 1/2	29 1/2
PROJECTILE W/PLASTIC ROTATING BAND (HI-FRAG)	29 1/4	30 1/4



DETAIL C

CENTER GATE (1 LAYER)
2 REQUIRED

SEE TABLE VII

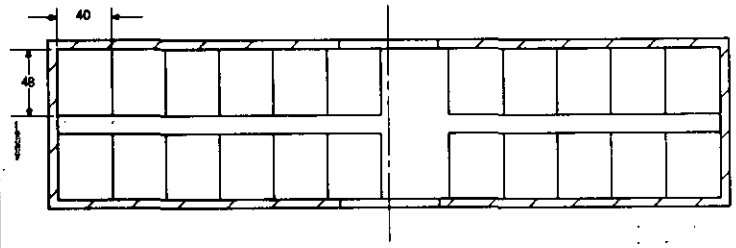


DETAIL D

DOORWAY PROTECTION
2 REQUIRED

WHEN BOXCAR HAS STEEL DOOR POSTS, DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) PAGE 46, FIG. 21 IS REQUIRED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.



LOADING PLAN
(SINGLE LAYER - 22 UNIT LOADS)

CARLOAD DATA

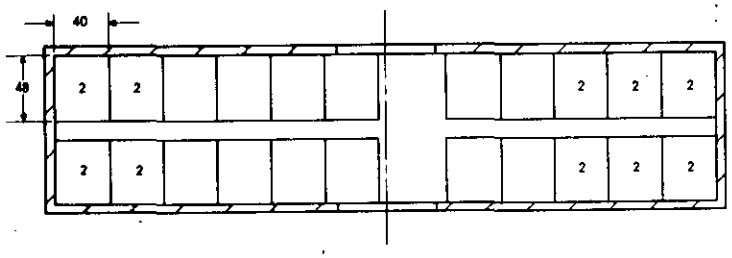
NUMBER OF UNIT LOADS 22
CARLOAD WEIGHT ..... SEE TABLE VIII

TABLE VIII

ITEM	CARLOAD WEIGHT (LBS) PALLET ADAPTER MK 11	
	MOD 0	MOD 1
PROJECTILE W/O WATERPROOF PROTECTING CAP	81641	80410
PROJECTILE W/WATERPROOF PROTECTING CAP	83754	82522
ROCKET ASSISTED PROJECTILE MK 58 MOD 0	76362	75130
PROJECTILE W/PLASTIC * ROTATING BAND (HI-FRAG)	80520	

INCLUDES 616 LBS (APPROX.) DUNNAGE WT.

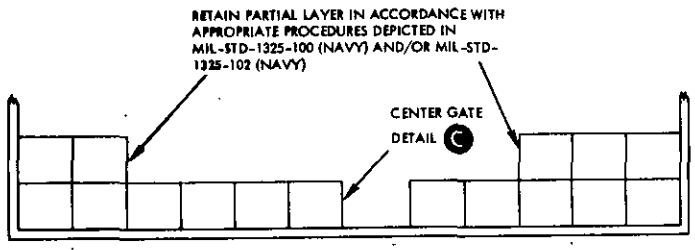
*U.L. HAS PALLET ADAPTER MK 11 MOD 2 BOTTOM AND MK 11 MOD 1 TOP



LOADING PLAN
(PARTIAL UPPER LAYER - 32 UNIT LOADS SHOWN)

CARLOAD DATA

NUMBER OF UNIT LOADS 32
CARLOAD WEIGHT ..... SEE TABLE IX



ELEVATION

TABLE IX

ITEM	CARLOAD WEIGHT (LBS) (APPROX) PALLET ADAPTER MK 11	
	MOD 0	MOD 1
PROJECTILE W/O WATERPROOF PROTECTING CAP	119456	117664
PROJECTILE W/WATERPROOF PROTECTING CAP	122528	120736
ROCKET ASSISTED PROJECTILE MK 58 MOD 0	111776	109984
*PROJECTILE W/PLASTIC ROTATING BAND (HI-FRAG)	117824	

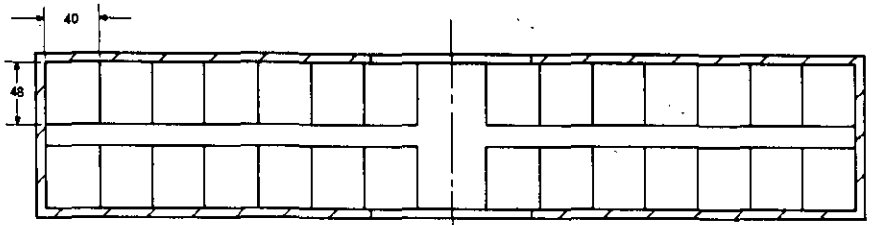
INCLUDES 1600 LBS (APPROX.) DUNNAGE WT.

*UNIT LOAD HAS PALLET ADAPTER MK 11 MOD 2 BOTTOM AND MK 11 MOD 1 TOP

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.

50 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD MAY CONSIST OF EITHER 40 UNIT LOADS WITH 28 UNIT LOADS IN THE FIRST LAYER AND A PARTIAL LAYER OF 12 UNIT LOADS IN THE SECOND LAYER (48" DIMENSION OF U.L. CROSSWISE IN THE CAR), OR 28 UNIT LOADS COMPRISING THE FIRST LAYER ONLY. THE PARTIAL LAYER SHALL BE DUNNAGED IN ACCORDANCE WITH THE PROCEDURES DEPICTED IN THIS DOCUMENT WHEN LOADING BOXCARS WITH WOOD SIDEWALLS. WHEN LOADING PARTIAL LAYERS WITH OTHER THAN THE QUANTITY OF UNIT LOADS SHOWN OR WHEN LOADING PARTIAL LAYERS IN BOXCARS WITH METAL SIDEWALLS, DUNNAGING SHALL BE IN ACCORDANCE WITH THE PROCEDURES DEPICTED IN MIL-STD-1325-100 (NAVY) OR MIL-STD-1325-102 (NAVY) RESPECTIVELY. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. QUANTITIES CAN ALSO BE ADJUSTED BY SUBSTITUTING DUMMY UNIT LOADS, (SEE PAGE 14) AS REQUIRED TO COMPLY WITH THE LOAD LIMIT OF THE BOXCAR.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100 OR PARTIAL UPPER LAYER BRACING IN ACCORDANCE WITH MIL-STD-1325-102 (NAVY). SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH, SEE MIL-STD-1325 (NAVY).
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS, AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR, AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICENT LENGTH TO SPAN DOORS IS NOT AVAILABLE, RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED, BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED. STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4 " STEEL STRAPS PER STACK, EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS AND A TOP SWAY BRACE CONSISTING OF A SUPPORT PIECE AND A CROSS PIECE, TWO PER STACK (SEE DETAIL D) MUST BE INSTALLED BETWEEN THE ROWS OF THE DOORWAY STACKS. STRAP IS SECURED TO SUPPORT PIECE WITH TWO 1 1/4 " STAPLES. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.



LOADING PLAN
(SINGLE LAYER - 28 UNIT LOADS)

CARLOAD DATA
NUMBER OF UNIT LOADS 28
CARLOAD WEIGHT ..... SEE TABLE X

TABLE X

ITEM	CARLOAD WEIGHT (LBS) (APPROX) PALLET ADAPTER MK 11	
	MOD 0	MOD 1
PROJECTILE W/O WATERPROOF PROTECTIVE CAP	103,871	102,303
PROJECTILE W/WATERPROOF PROTECTIVE CAP	106,559	104,991
ROCKET ASSISTED PROJECTILE MK 58 MOD 0	97,151	95,583
*PROJECTILE W/PLASTIC ROTATING BAND (HI-FRAG)	102,443	

INCLUDES 747 LBS (APPROX) DUNNAGE WT.
*UNIT LOAD HAS PALLET ADAPTER MK 11 MOD 2 BOTTOM AND MK 11 MOD 1 TOP.

* USE SWAY BRACE ASSEMBLY, DETAIL A, FOR STEEL FLOOR CARS.
** W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.
*** 2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

PIECE NO.	DESCRIPTION	SIZE	NO. PCS. REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
25	UPPER WALL CLEAT	2 x 4 x 18	4	CAR WALL	4	10d
24	INTERMEDIATE WALL CLEAT	2 x 4 x 23	4	CAR WALL	4	10d
23	HORIZONTAL BACK-UP CLEAT	2 x 6 x EXTEND ACROSS DOOR	8	6	18	16d
22	DIAGONAL BRACE	4 x 4 x 70 3/8	8	16, 18	1 EA END	60d
21	HORIZONTAL POCKET CLEAT	2 x 6 x 24	8	19	7	16d
20	HORIZONTAL POCKET CLEAT	2 x 6 x 36	8	18	10	16d
19	HORIZONTAL WALL CLEAT	2 x 6 x EXTEND ACROSS DOOR	8	CAR WALL	40	10d
18	CENTER CLEAT	2 x 4 x 36	4	15	7	16d
17	CROSS BRACE STIFFENER	2 x 6 x CAR WIDTH	4	15	2 PER FOOT	12d
16	CROSS BRACE	4 x 4 x CAR WIDTH	4	SEE 14	-	-
15	CROSS BRACE STIFFENER	2 x 6 x CAR WIDTH	4	3	2 PER FOOT	10d
14	LOWER WALL CLEAT	2 x 4 x 4	4	CAR WALL	3	10d

LIST OF MATERIALS AND NAILING DATA

PIECE NO.	DESCRIPTION	SIZE	NO. PCS. REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
13	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	4	11	2 PER JOINT	10d
12	VERTICAL DOORWAY MEMBER	2 x 3 x 42	4	DOOR POST	2 PER JOINT	20d
11	STRUT	4 x 4 x *** WEDGE FIT	8	5	2 PER JOINT	16d
10	HOLD DOWN CLEAT	2 x 3 x CUT TO FIT	4	8	5	16d
9	HOLD DOWN CLEAT	2 x 3 x CUT TO FIT	4	7	5	10d
8	HOLD DOWN SPACER	2 x 4 x CAR WIDTH -1	2	5	3 PER JOINT	10d
7	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH -1	4	5	3 PER JOINT	10d
6	CENTER GATE VERTICAL	2 x 6 x 42	16	SEE 4	-	-
5	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH -1	4	5	3 PER JOINT	10d
4	CROSS BRACE	2 x 4 x W **	18	SEE 3	-	-
3	SWAY BRACE LONGITUDINAL	1 x 4 x 40	12	2, 4	3 PER JOINT	10d
2	SWAY BRACE SUPPORT	1 x 4 x (W + 8) **	12	SEE 3	-	-
1	SLEEPER	2 x 4 x 40 *	28	CAR FLOOR	2 PER FOOT	16d

LIST OF MATERIALS AND NAILING DATA

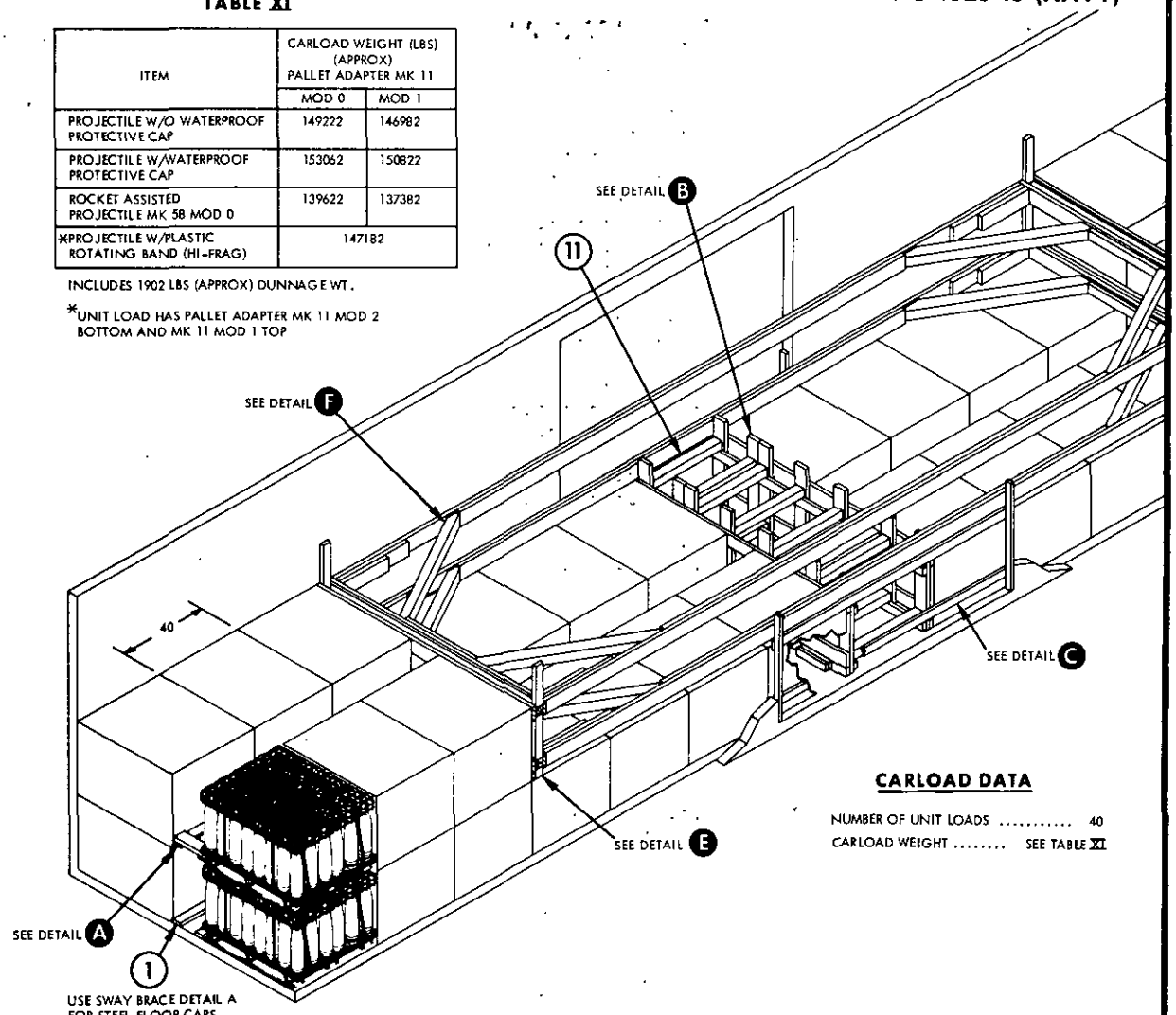
Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.

TABLE XI

ITEM	CARLOAD WEIGHT (LBS) (APPROX) PALLET ADAPTER MK 11	
	MOD 0	MOD 1
PROJECTILE W/O WATERPROOF PROTECTIVE CAP	149222	146982
PROJECTILE W/WATERPROOF PROTECTIVE CAP	153062	150822
ROCKET ASSISTED PROJECTILE MK 58 MOD 0	139622	137382
*PROJECTILE W/PLASTIC ROTATING BAND (HI-FRAG)	147182	

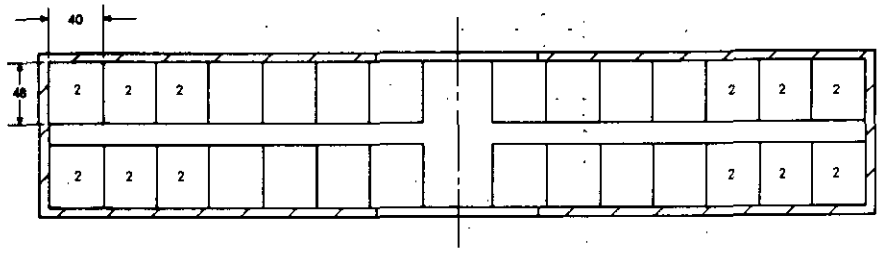
INCLUDES 1902 LBS (APPROX) DUNNAGE WT.

*UNIT LOAD HAS PALLET ADAPTER MK 11 MOD 2 BOTTOM AND MK 11 MOD 1 TOP

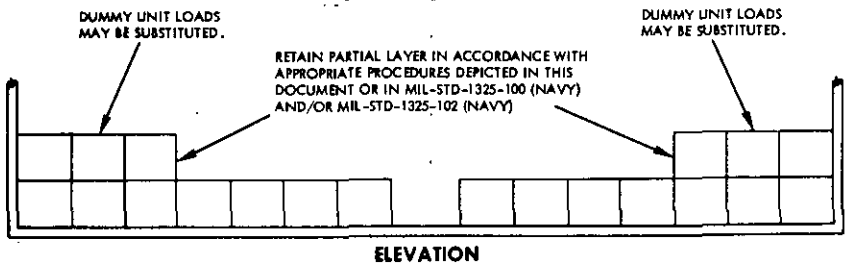


CARLOAD DATA

NUMBER OF UNIT LOADS 40
CARLOAD WEIGHT ..... SEE TABLE XI

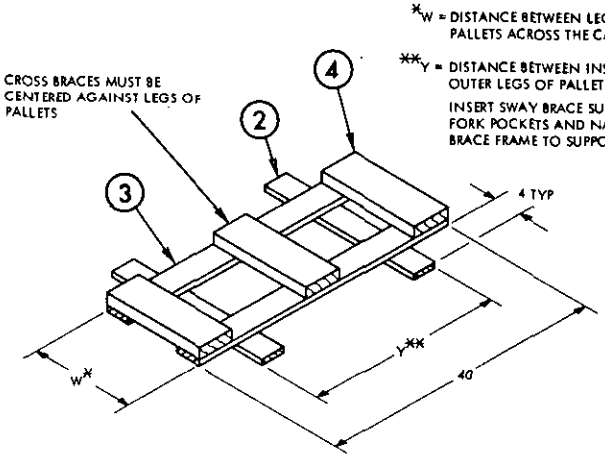


LOADING PLAN



ELEVATION

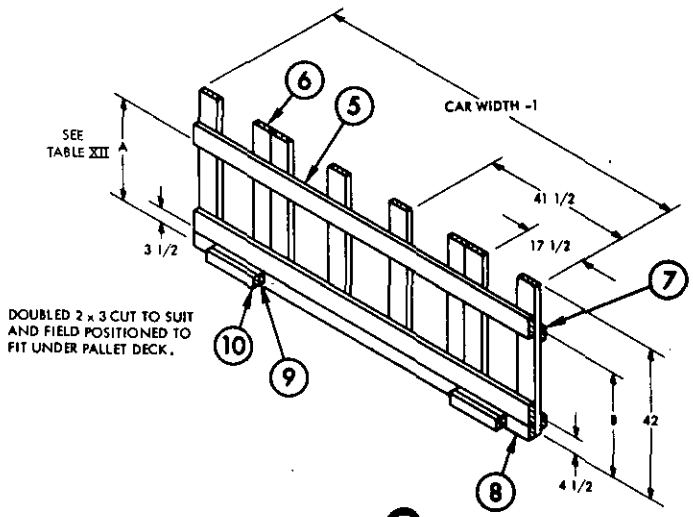
Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.



DETAIL A
SWAY BRACE ASSEMBLY
6 REQUIRED

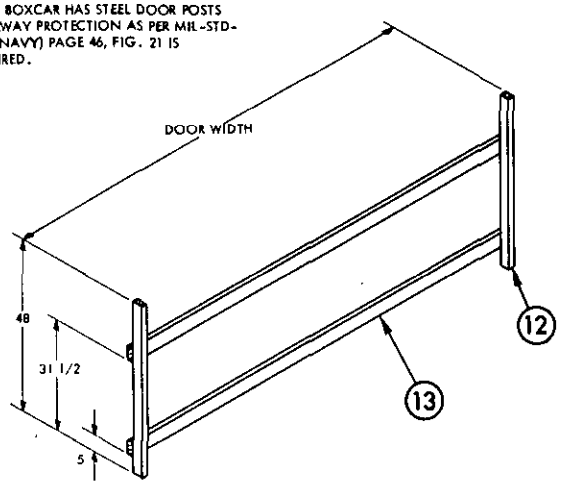
TABLE XII

ITEM	A	B
PROJECTILE W/O WATERPROOF PROTECTING CAP	26 1/4	27 1/4
PROJECTILE W/WATERPROOF PROTECTING CAP	28 1/4	29 1/4
ROCKET ASSISTED PROJECTILE MK 58 MOD 0	28 1/2	29 1/4
PROJECTILE W/PLASTIC ROTATING BAND (HI-FRAG)	29 1/2	30 1/4

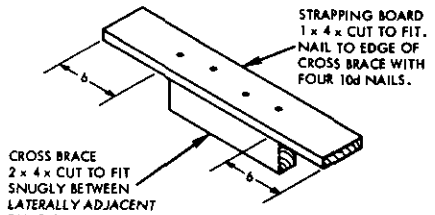


DETAIL B
CENTER GATE (1 LAYER)
2 REQUIRED
SEE TABLE XII

WHEN BOXCAR HAS STEEL DOOR POSTS DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) PAGE 46, FIG. 21 IS REQUIRED.

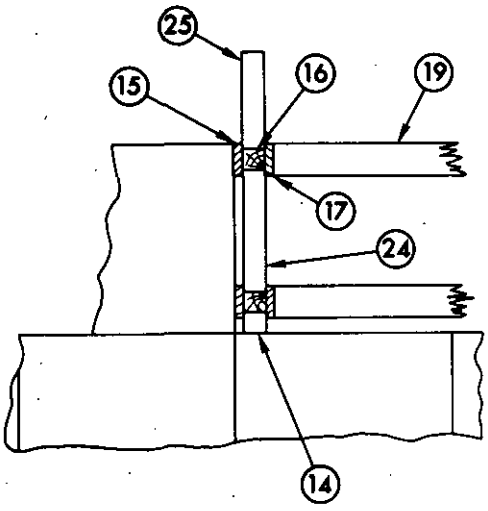


DETAIL C
DOORWAY PROTECTION
2 REQUIRED

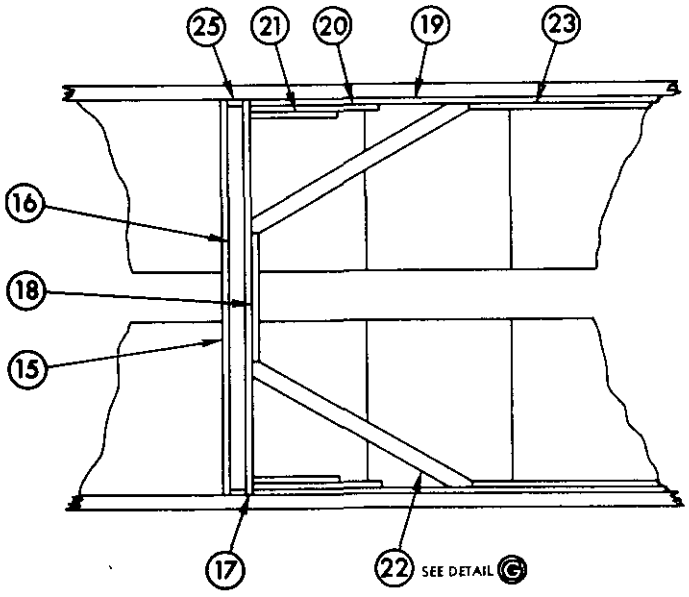


DETAIL D
TOP SWAY BRACE

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.

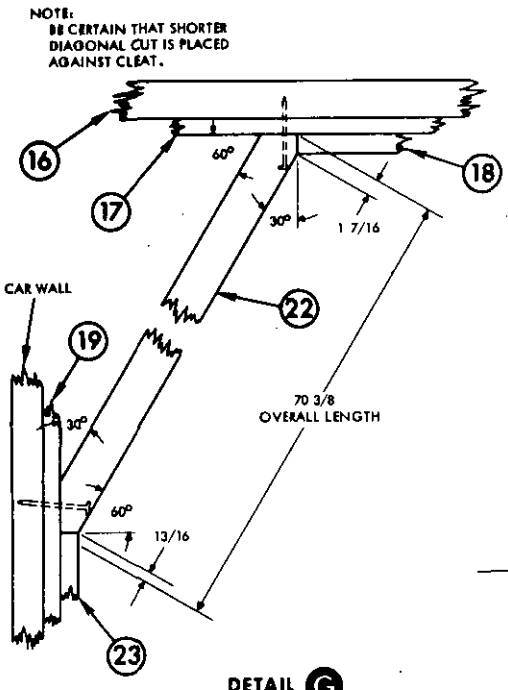


DETAIL E



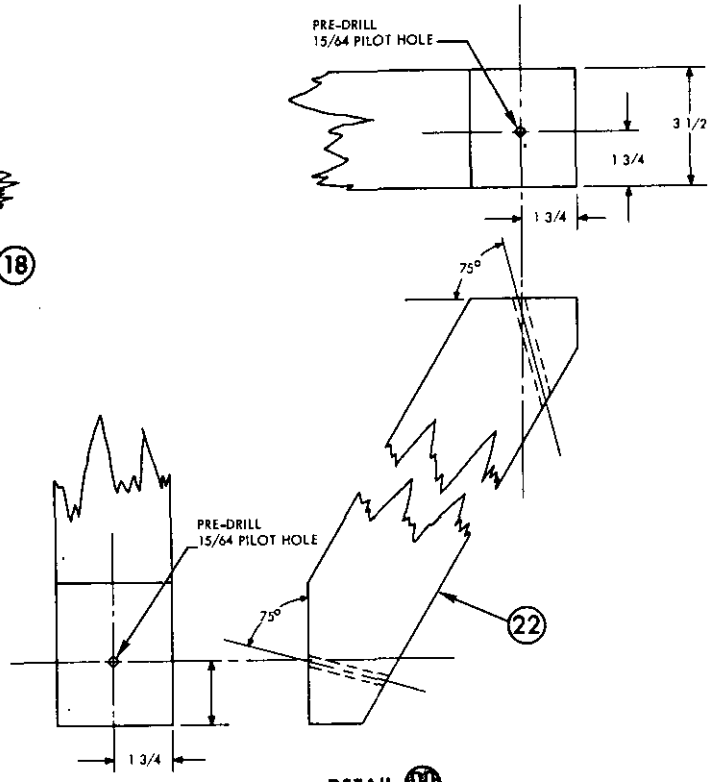
DETAIL F

PARTIAL LAYER BRACING



DETAIL G

POSITIONING O+ DIAGONAL BRACE



DETAIL H

DIAGONAL BRACE
LOCATION OF PILOT HOLE
FOR 60d NAIL

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.

DUMMY UNIT LOAD

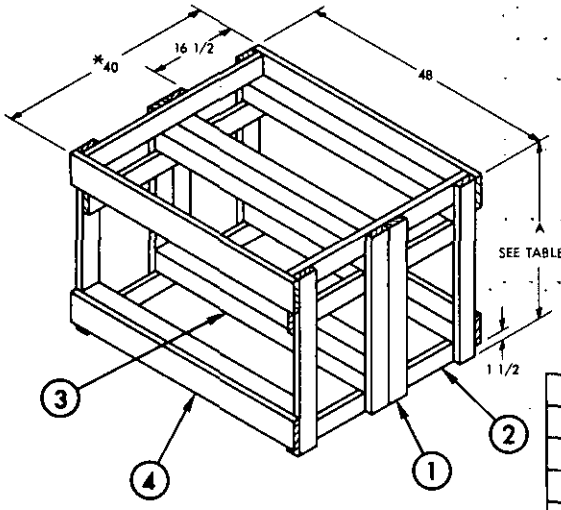


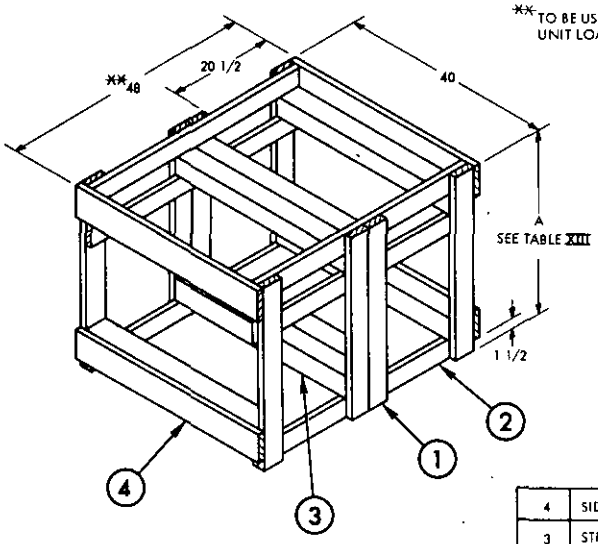
TABLE XIII

ITEM	A
PROJECTILE W/O WATERPROOF PROTECTING CAP	31 3/4
PROJECTILE W/WATERPROOF PROTECTING CAP	33 3/4
ROCKET ASSISTED PROJECTILE MK 58 MOD 0	34
PROJECTILE W/PLASTIC *ROTATING BAND (HI-FRAG)*	34 3/4

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER	SIZE
					NAILS	
4	SIDE LONGITUDINAL	2 x 6 x 48	4	1 & 2	3 PER JOINT	10d
3	STRUT	4 x 4 x 45	8	SEE 1	-	-
2	CROSS BRACE	2 x 4 x 37	6	SEE 1	-	-
1	END VERTICAL	2 x 4 x SEE TABLE "A"	8	3 2	2/JOINT 3/JOINT	16d 10d
LIST OF MATERIALS AND NAILING DATA						

*TO BE USED WHEN THE 48" DIMENSION OF THE UNIT LOAD IS LENTHWISE IN THE CAR.

**TO BE USED WHEN THE 48" DIMENSION OF THE UNIT LOAD IS CROSSWISE IN THE CAR.



PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER	SIZE
					NAILS	
4	SIDE LONGITUDINAL	2 x 6 x 40	4	1 & 2	3 PER JOINT	10d
3	STRUT	4 x 4 x 37	8	SEE 1	-	-
2	CROSS BRACE	2 x 4 x 45	6	SEE 1	-	-
1	END VERTICAL	2 x 4 x SEE TABLE "A"	8	3 2	2/JOINT 3/JOINT	16d 10d
LIST OF MATERIALS AND NAILING DATA						

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

COMMANDING OFFICER
NAVAL WEAPONS STATION EARLE
NAVAL WEAPONS HANDLING LABORATORY (803)
COLTS NECK, NEW JERSEY 07722

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.